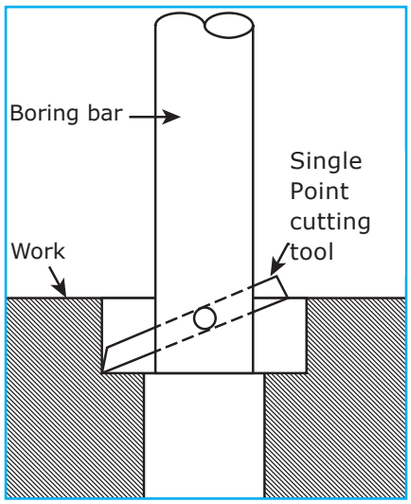
The figure illustrates the boring operation.

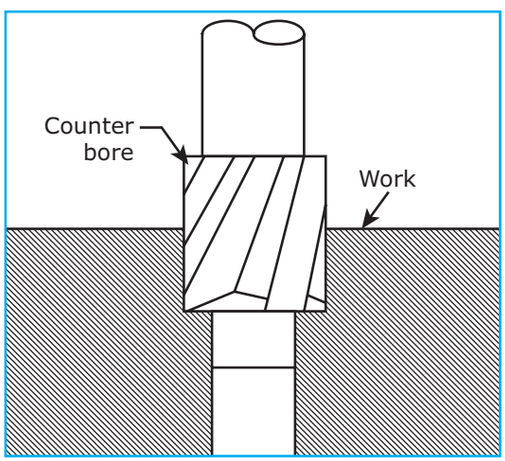


Boring

Counter boring

Counter boring is the operation of enlarging the end of the hole cylindrically. The enlarged hole forms a square shoulder with the original hole. This is necessary in some cases to accommodate the heads of bolts, studs and pins. The tool used for counter boring is known as counter bore.

The counter bores are made with cutting edges which may be straight (or) spiral. The cutting speed for counter boring is at least 25% lesser than that of drilling.

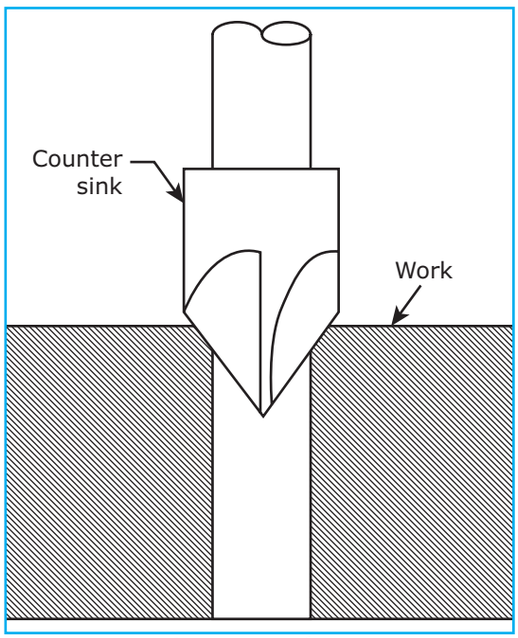


Counter Boring

Counter Sinking

Counter sinking is the operation of making a cone shaped enlargement at the end of the hole. The included angle of the conical surface may be in the range of 60^0 82^0 (or) 90^0. It is used to provide recess for a flat headed screw or a counter sunk rivet fitted into the hole. The tool used for counter sinking is known as a counter sink. It has multiple cutting edges on its conical surface. The cutting speed for counter sinking is 25% lesser than that of drilling.

The figure illustrates counter sinking operation.



Counter sinking

Spot facing

Spot facing is the operation of smoothing and squaring the surface around a hole. It is done to provide proper seating for a nut (or) the head of screw. A counter bore or a special spot facing tool may be employed for this purpose.

The following figure illustrates spot facing operation.

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